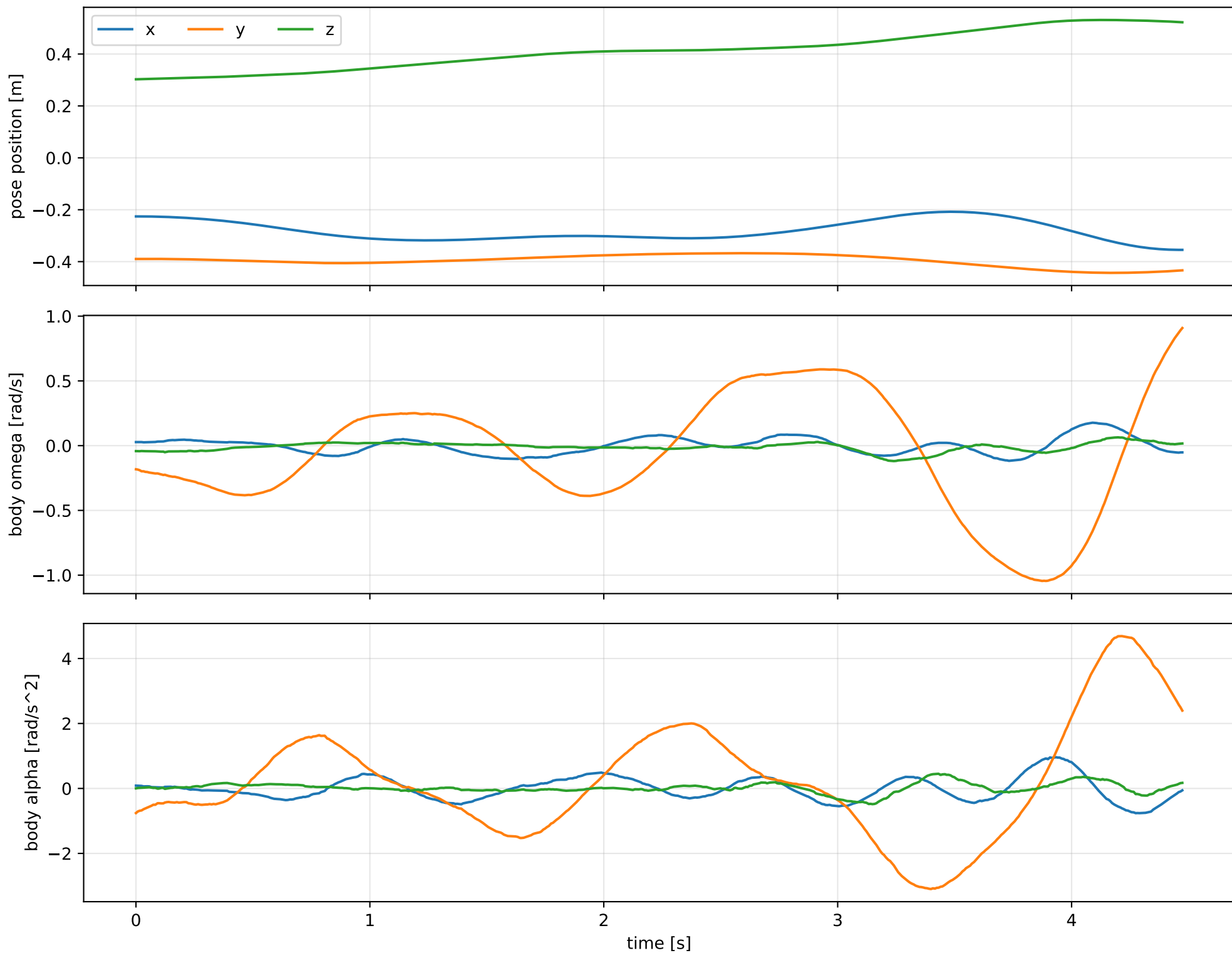
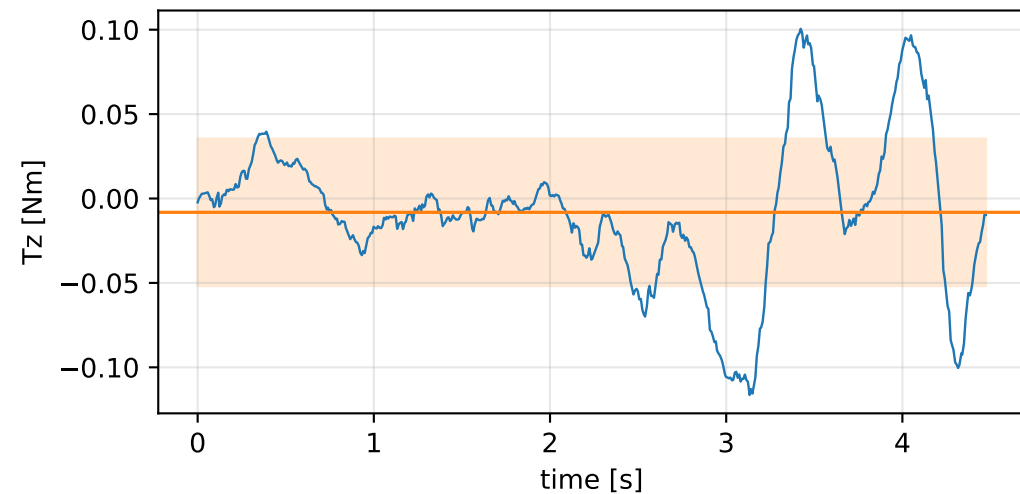
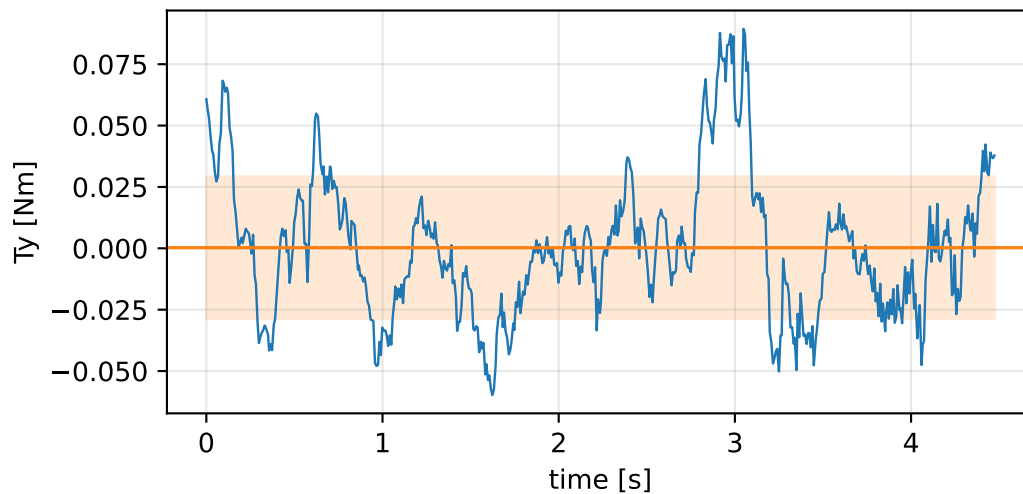
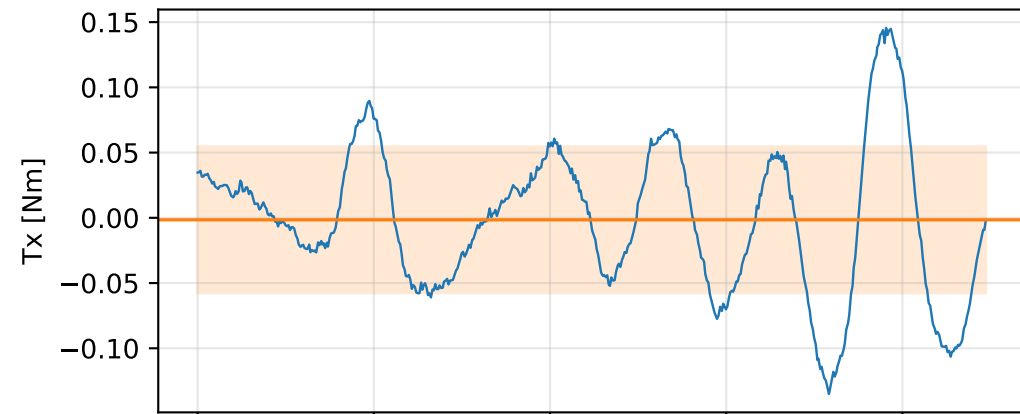
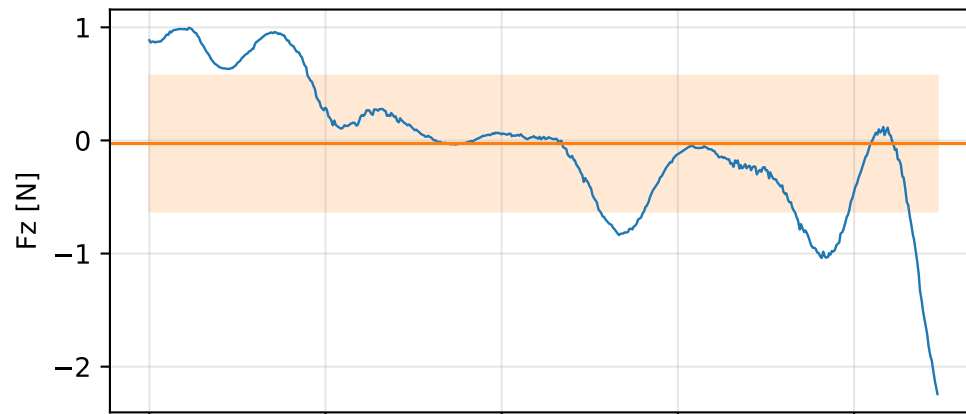
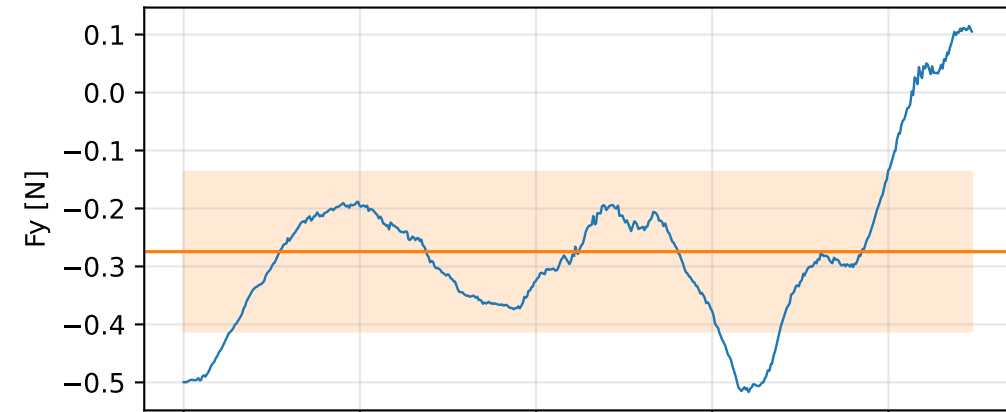
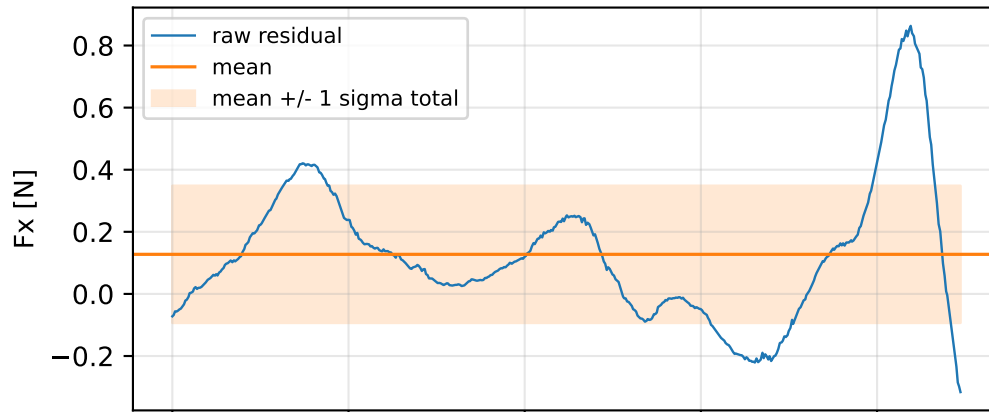


inertia_over_mass_yy_nominal: SG trajectory and derived motion



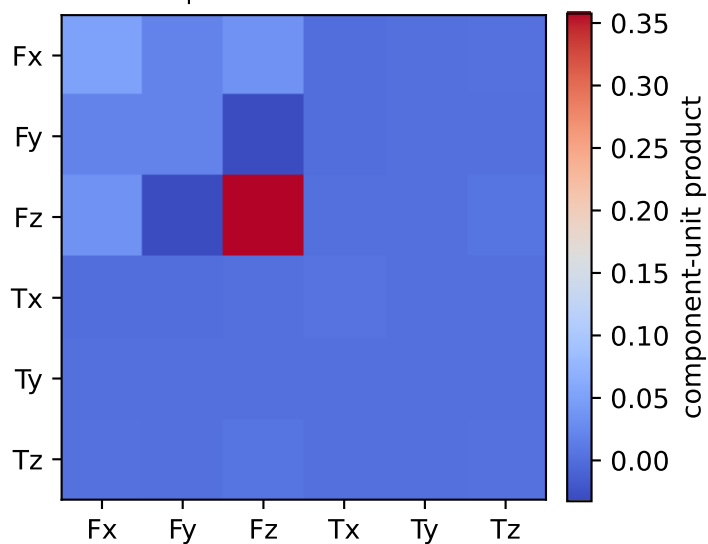
inertia_over_mass_yy_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



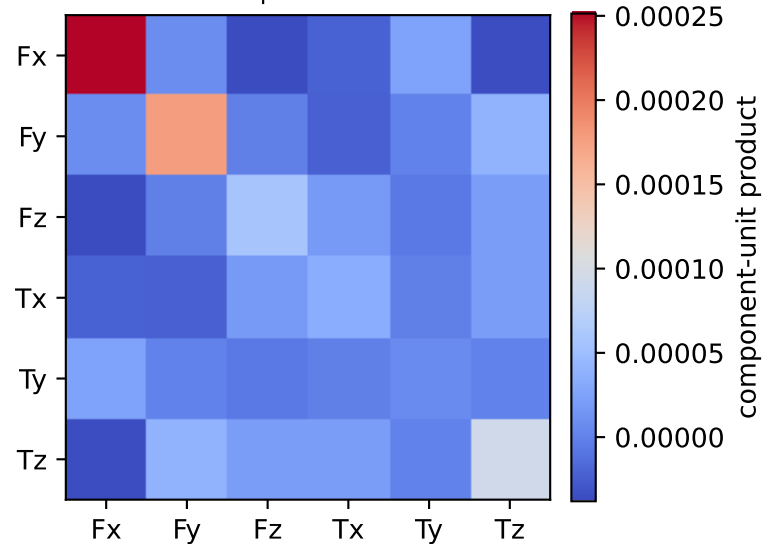
vector RMS about zero: force 0.71875 N, torque 0.077046 Nm; component std about mean: force [0.2206 0.1371 0.598], torque [0.0561 0.0289 0.0436]

inertia_over_mass_yy_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.002 0.005 0.057 0.364]

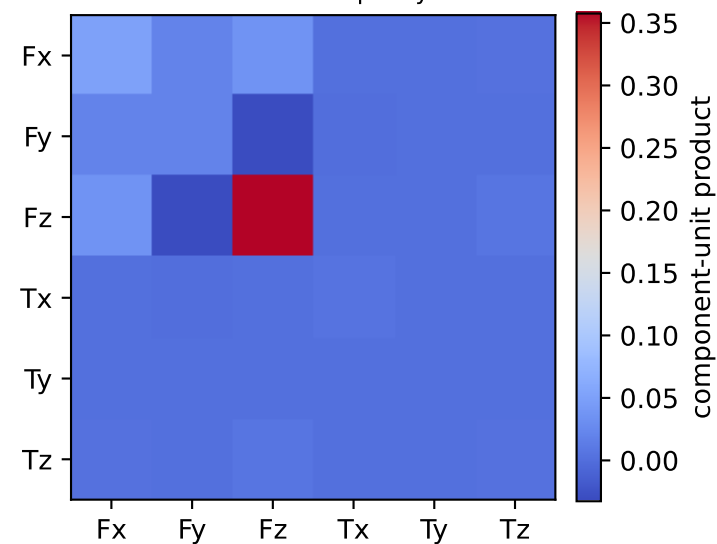
empirical total covariance



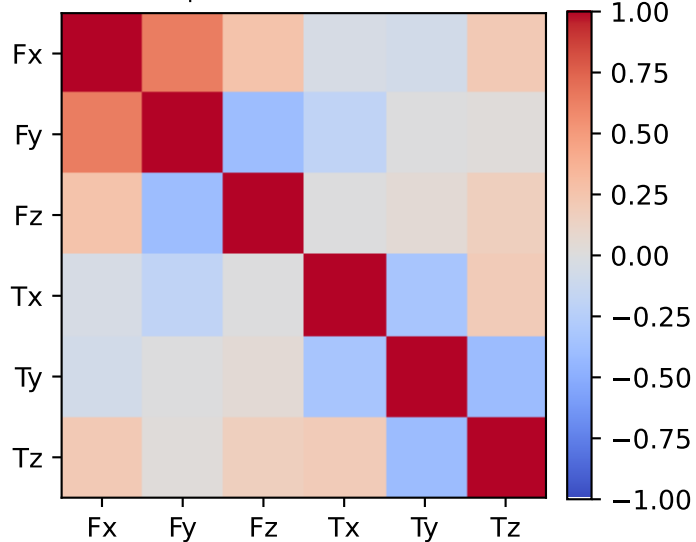
mean full-SG prediction covariance



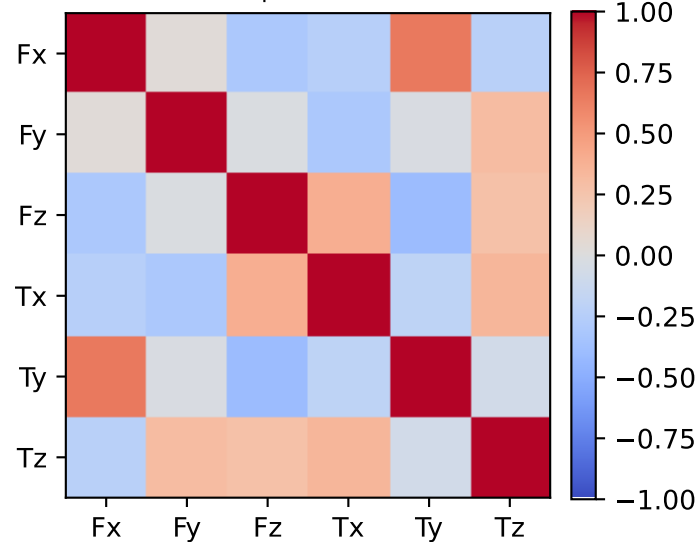
PSD excess model discrepancy covariance



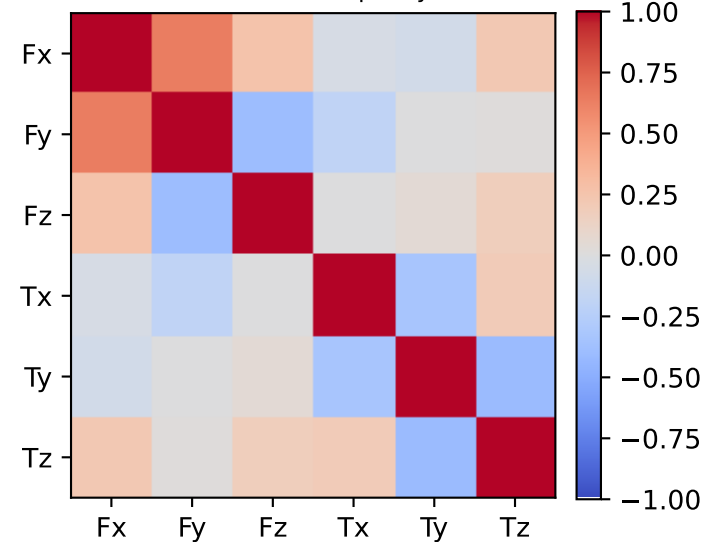
empirical total correlation



mean full-SG prediction correlation

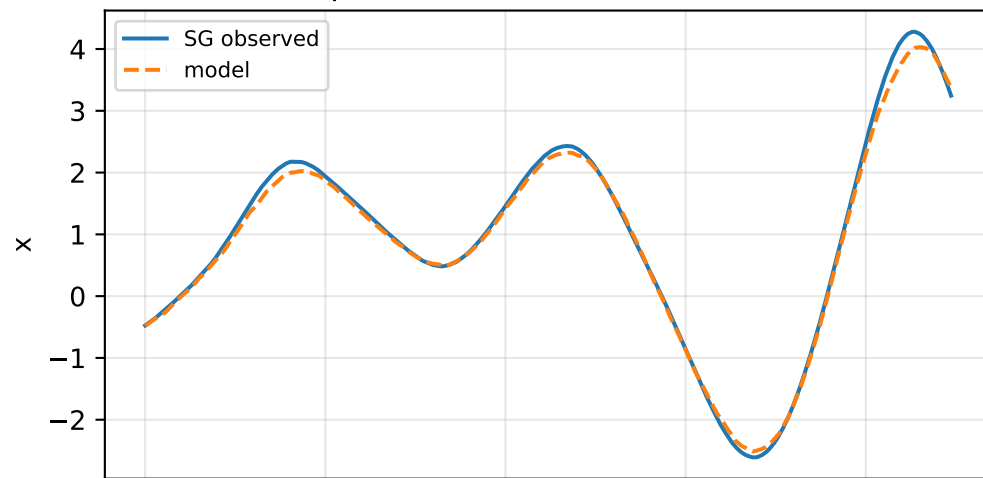


PSD excess model discrepancy correlation

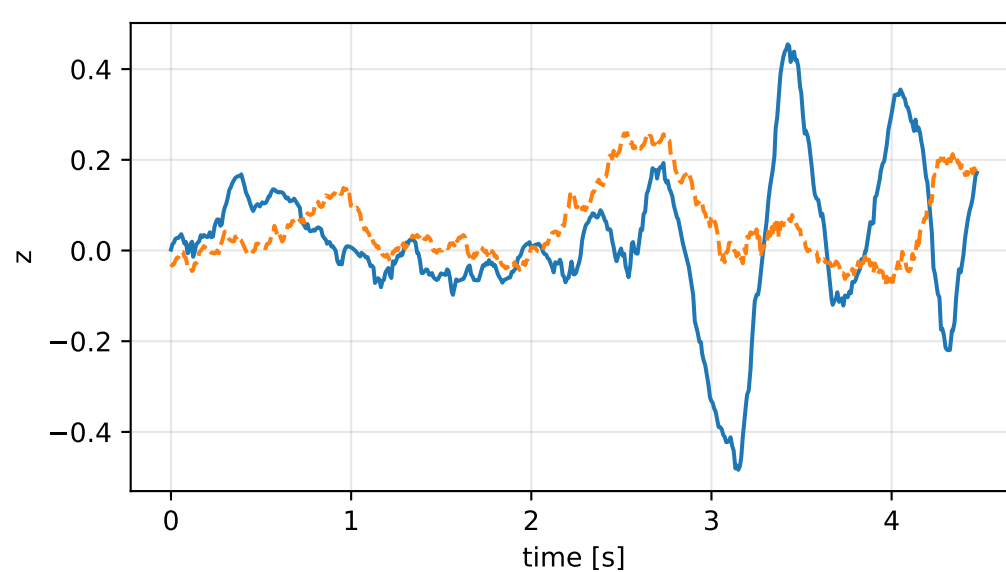
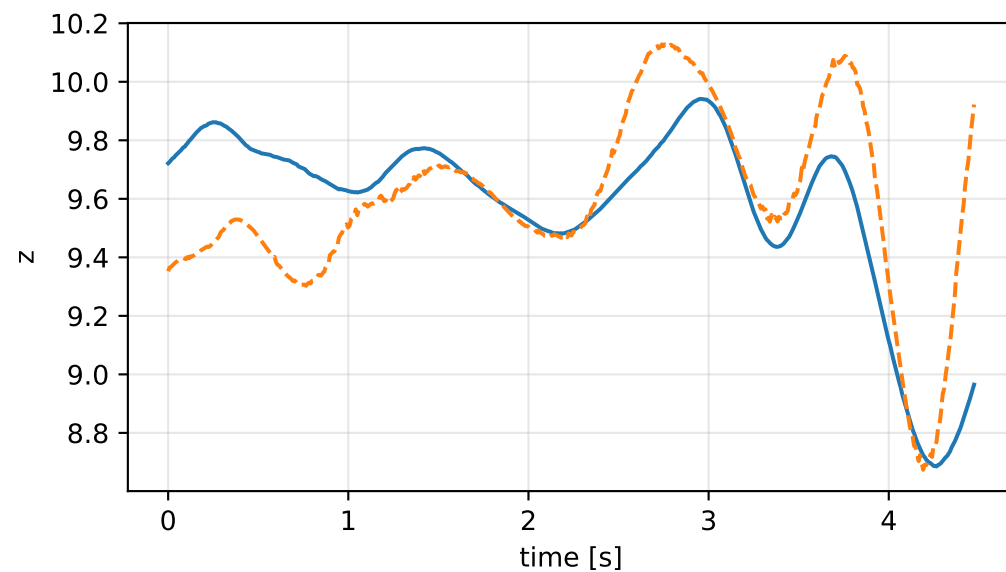
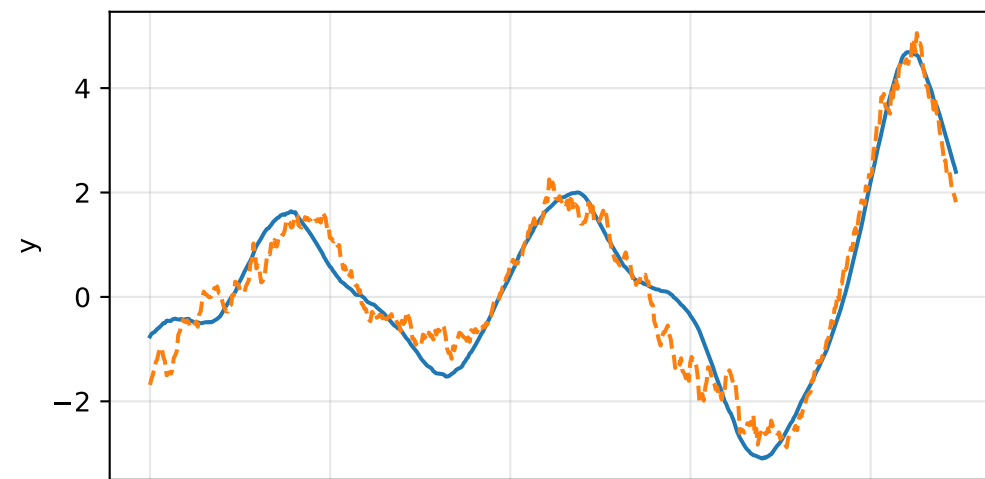
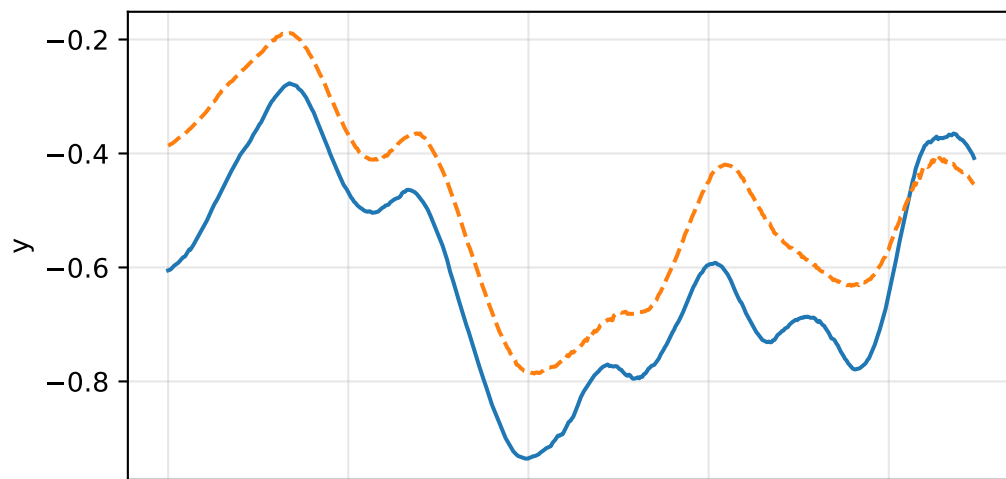
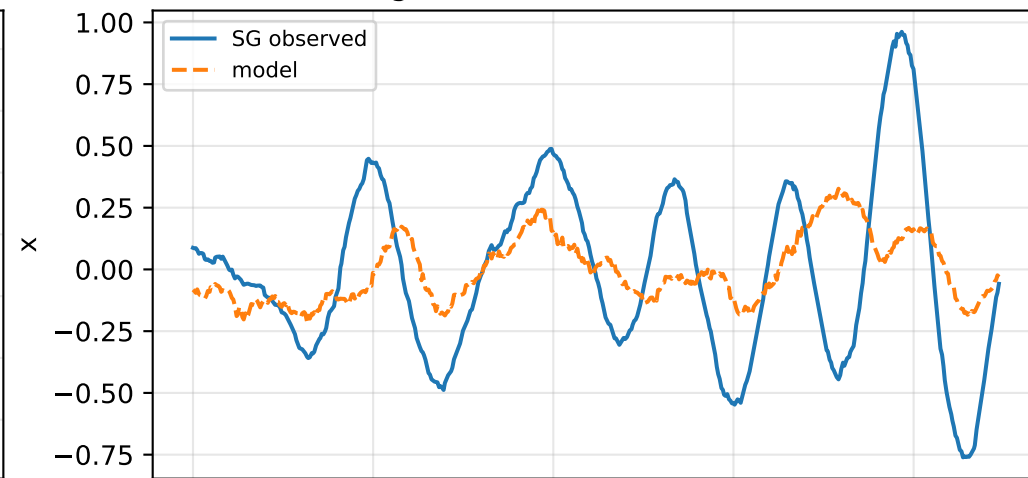


inertia_over_mass_yy_nominal: acceleration residual objective

specific acceleration [m/s^2]

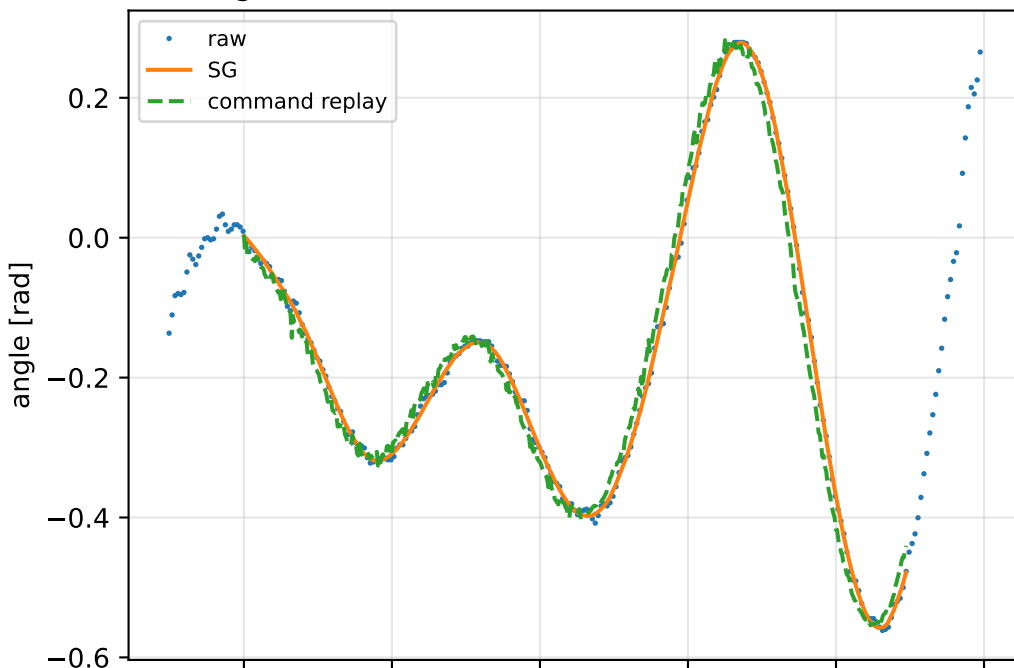


angular acceleration [rad/s^2]

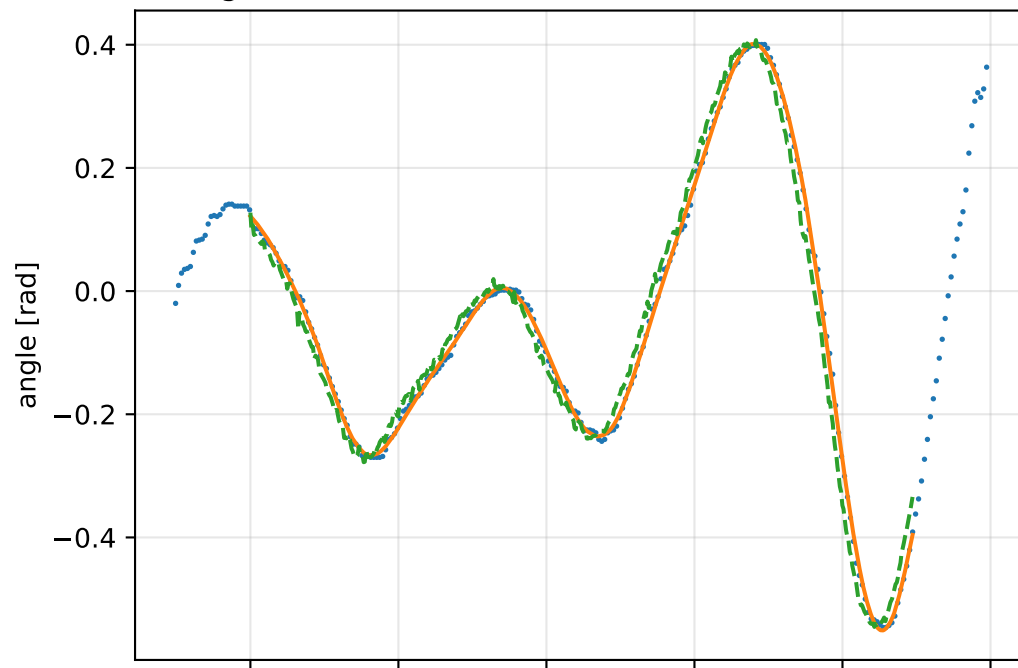


inertia_over_mass_yy_nominal: gimbal measurement audit (objective=measured_sg)

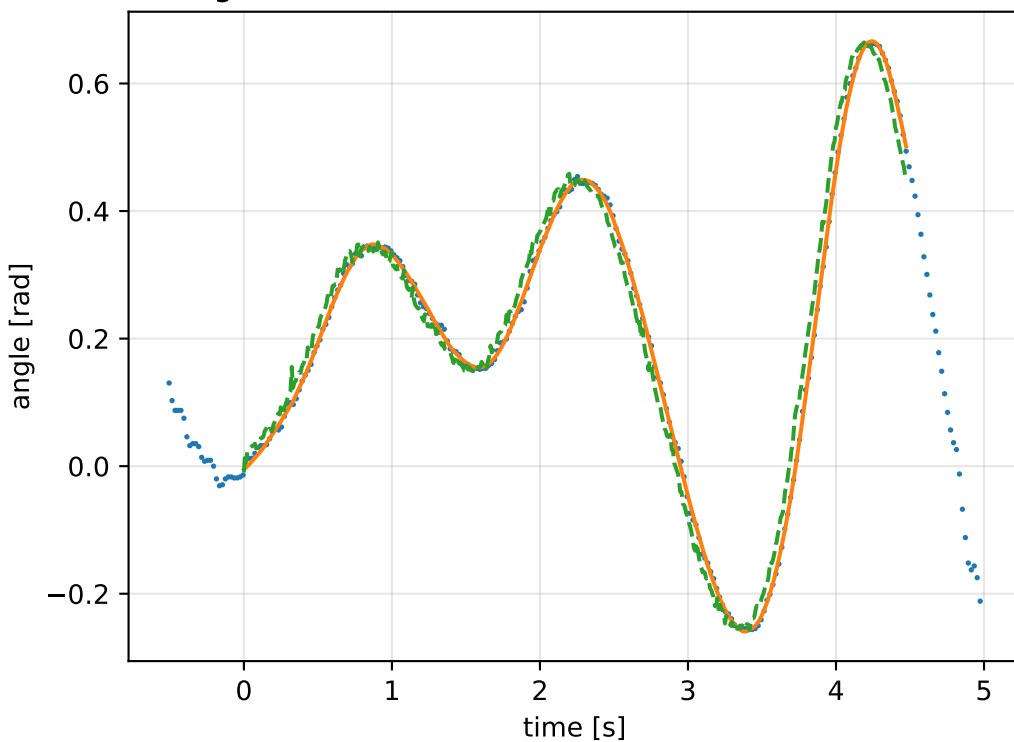
gimbal 1: RMSE=0.027, max=0.07529 rad



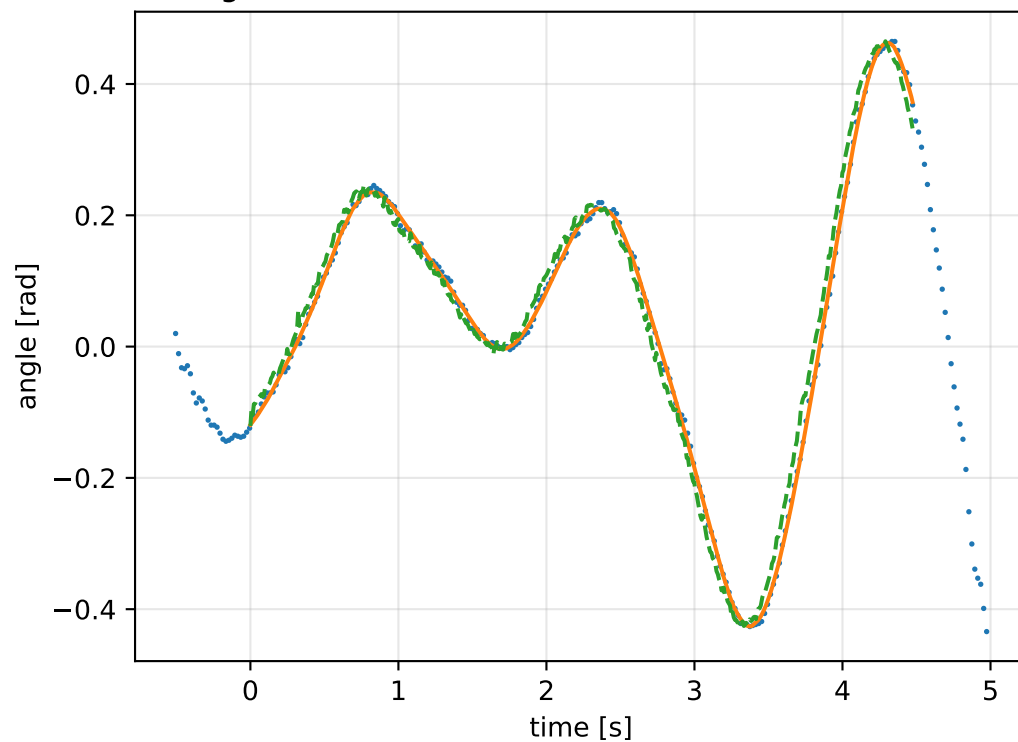
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

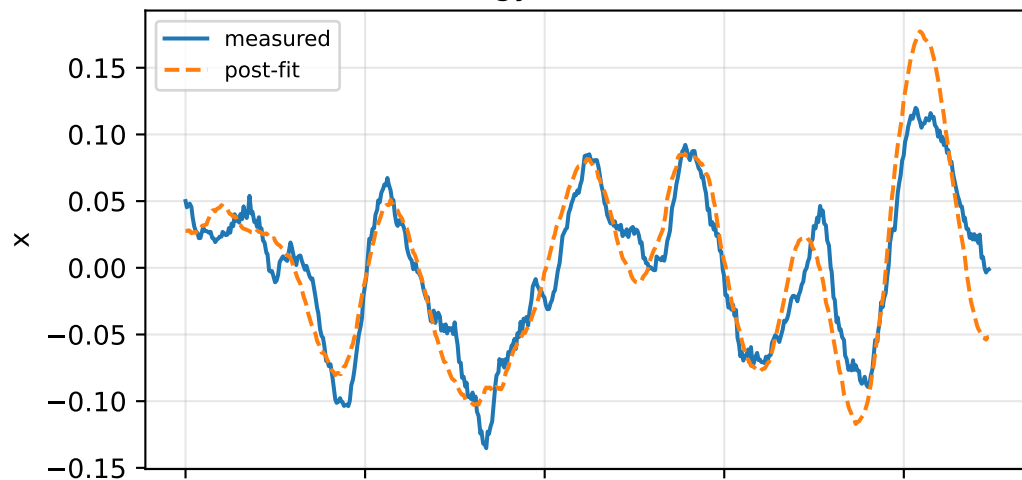


gimbal 4: RMSE=0.02598, max=0.07261 rad

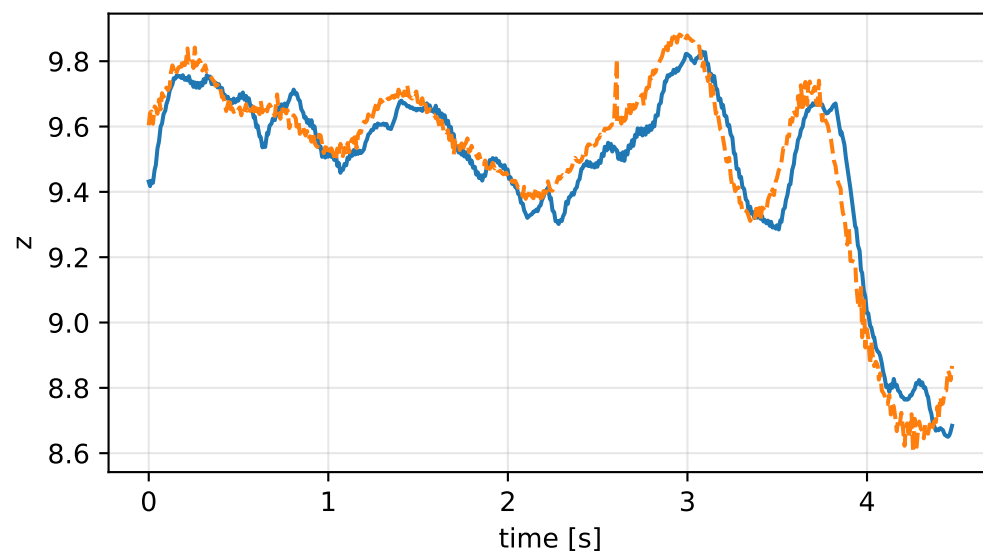
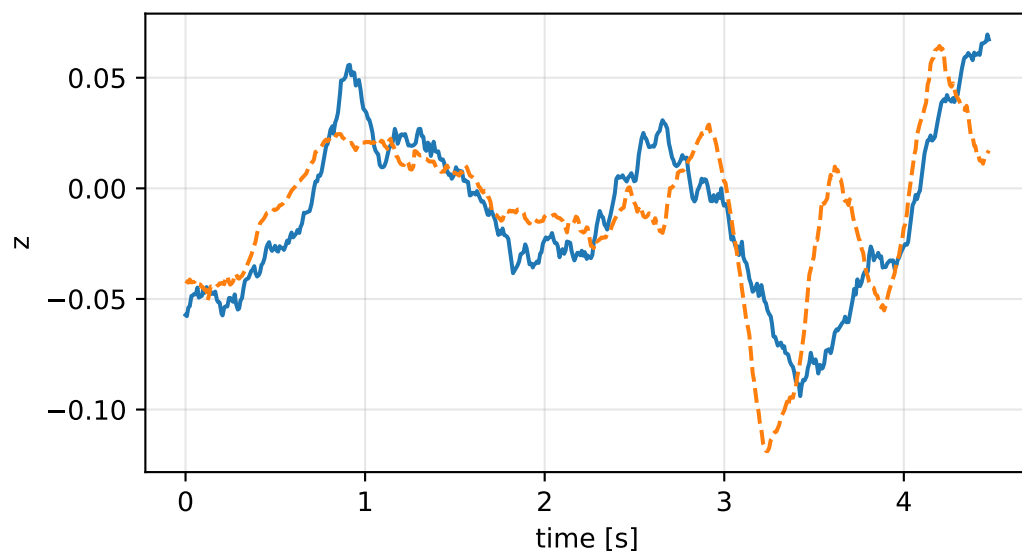
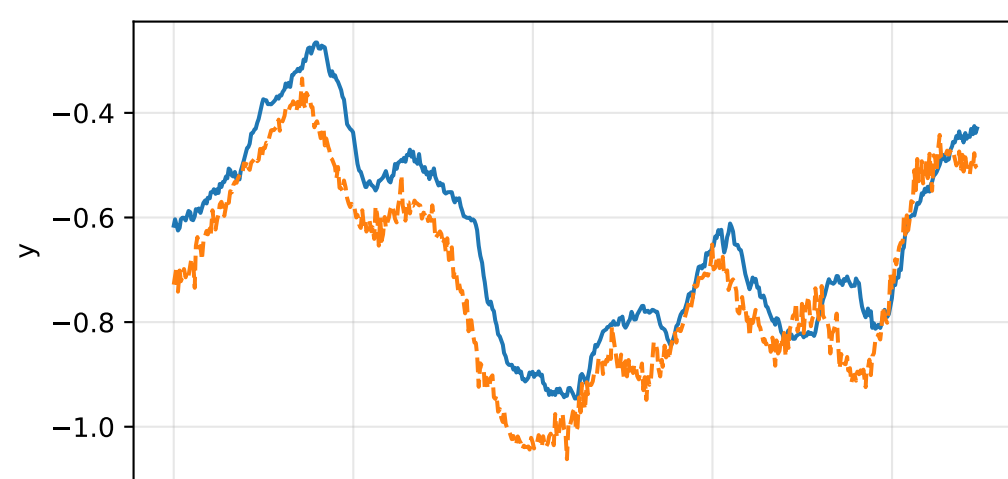
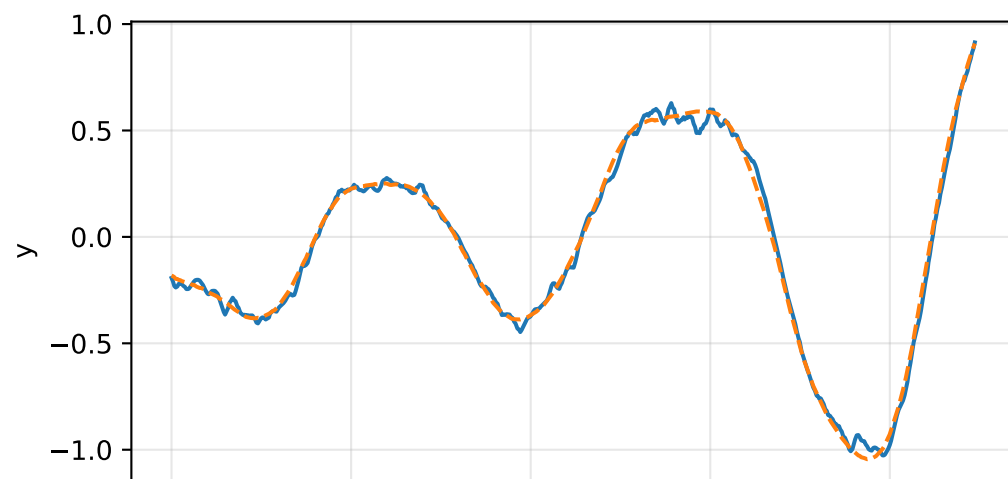
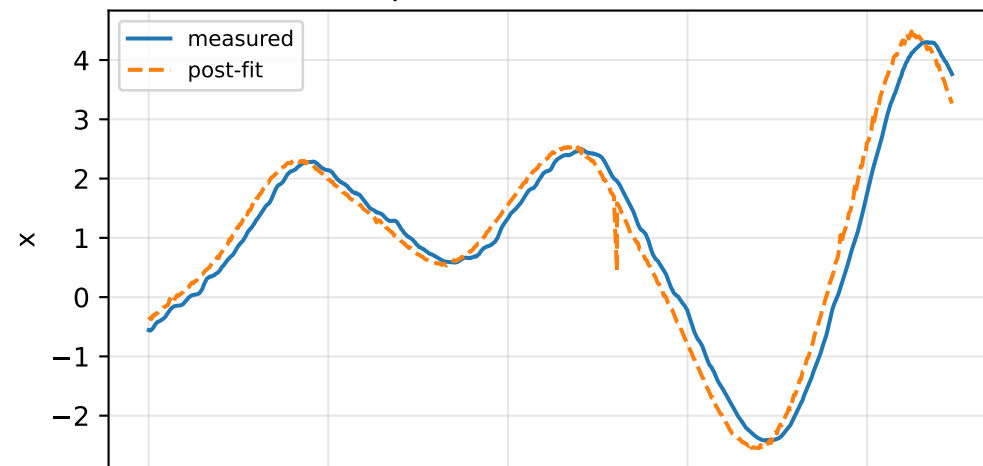


inertia_over_mass_yy_nominal: IMU comparison (diagnostic only)

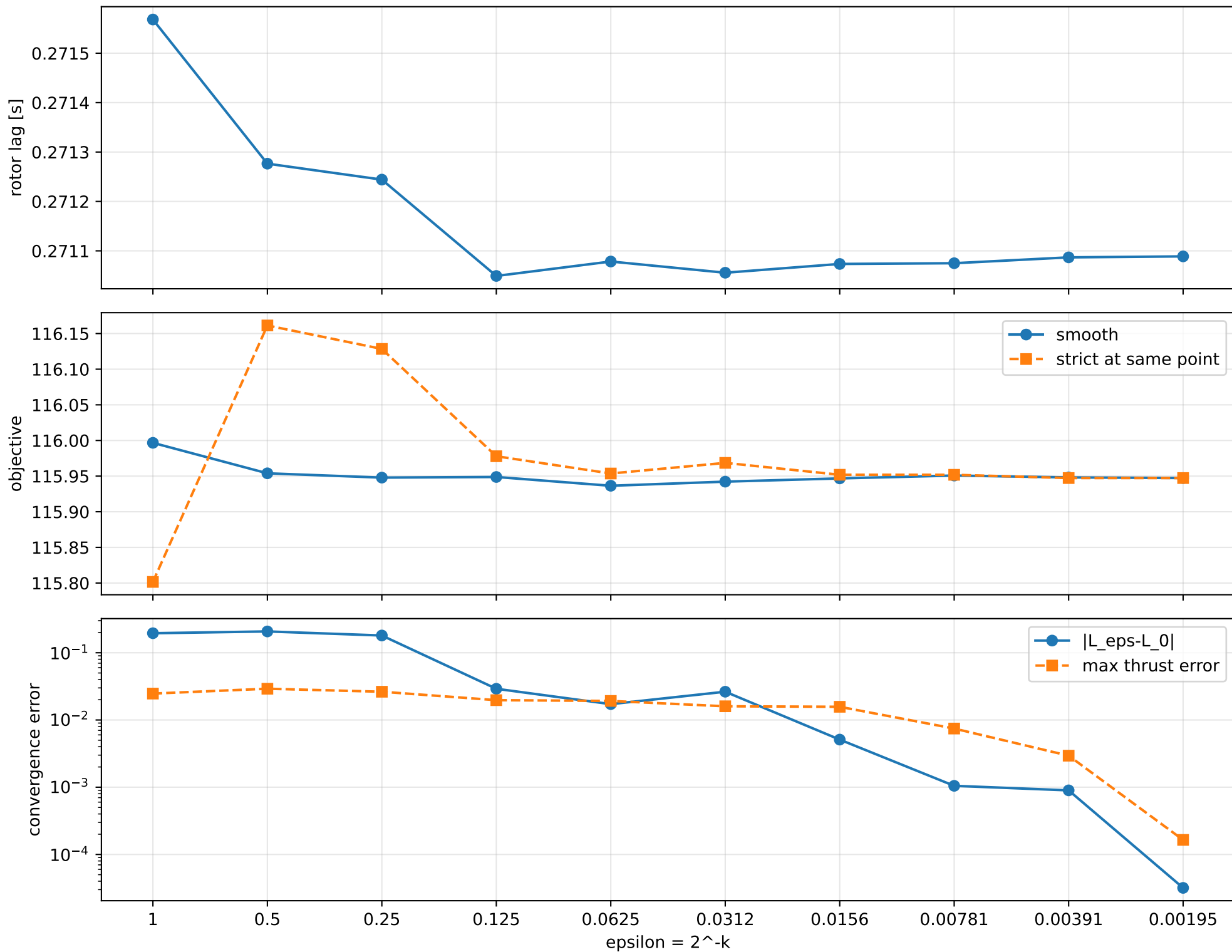
gyro [rad/s]



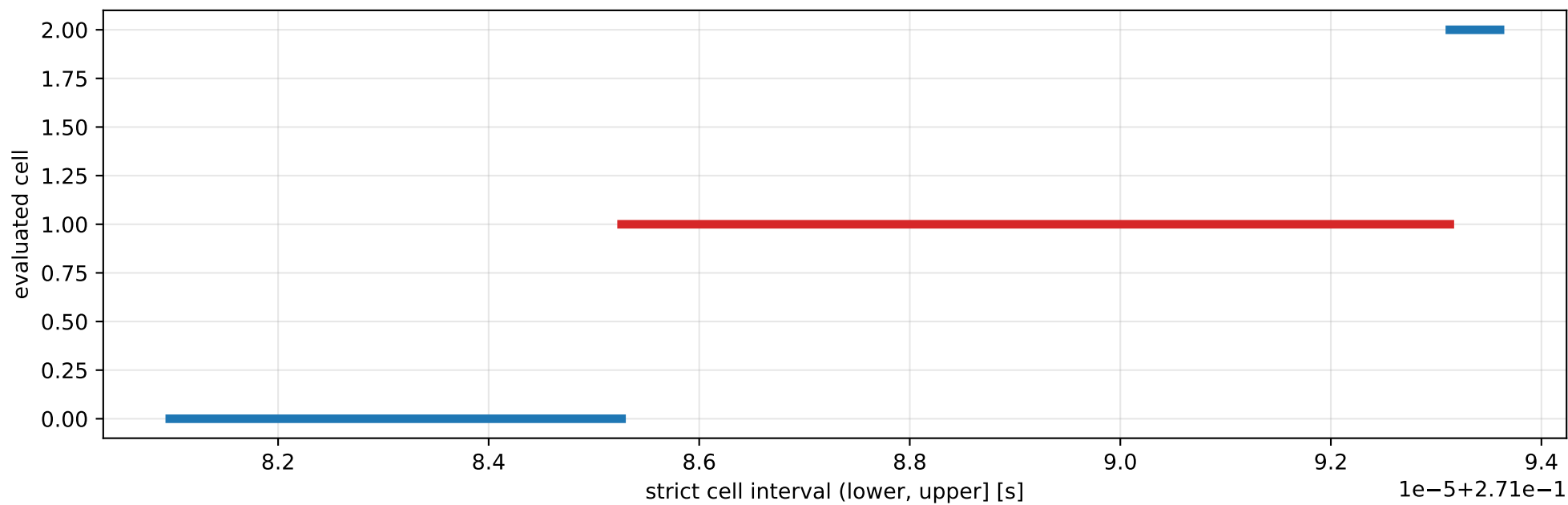
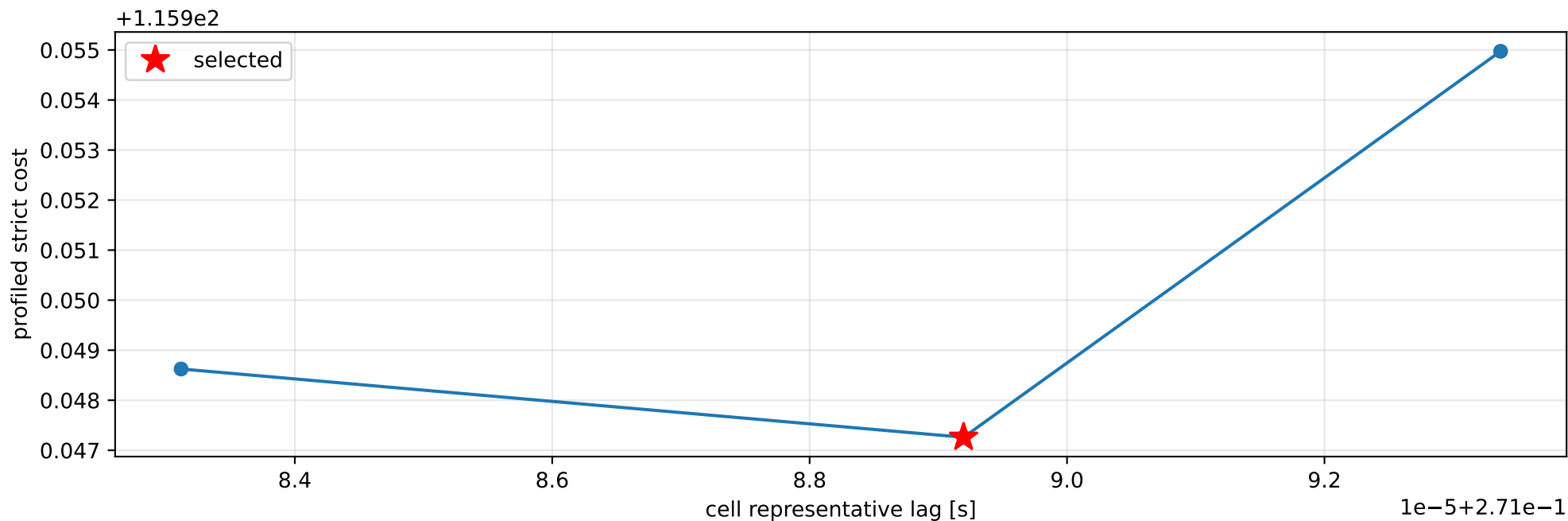
specific force [m/s²]



inertia_over_mass_yy_nominal: smooth-to-strict continuation



inertia_over_mass_yy_nominal: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271088801 s; data boundary=False

inertia_over_mass_yy_nominal: parameter estimate

Case: inertia_over_mass_yy_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.7655158

inertia [kg m²]:

```
[[ 0.20690431  0.00630048 -0.00353236]
 [ 0.00630048  0.07638535 -0.01372783]
 [-0.00353236 -0.01372783  0.28081244]]
```

CoG body [m]: [-0.02449034 0.00464419 0.01666246]

force effectiveness: [1.06865043 0.89740075 0.91285915 1.10591352]

scale-free J/m [m²]:

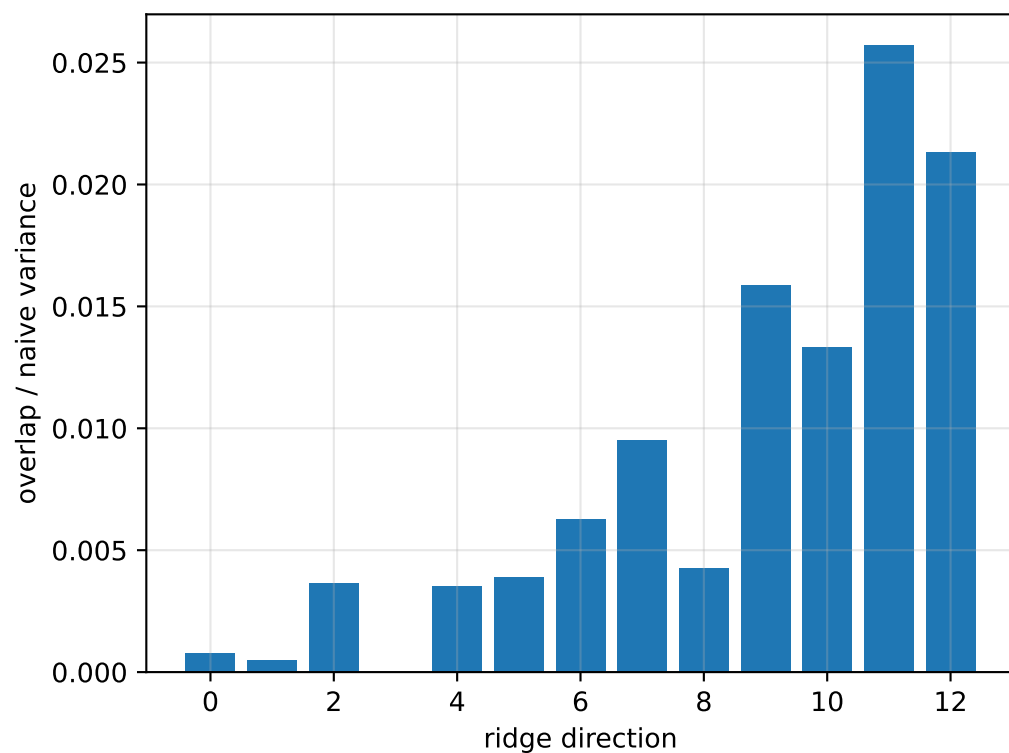
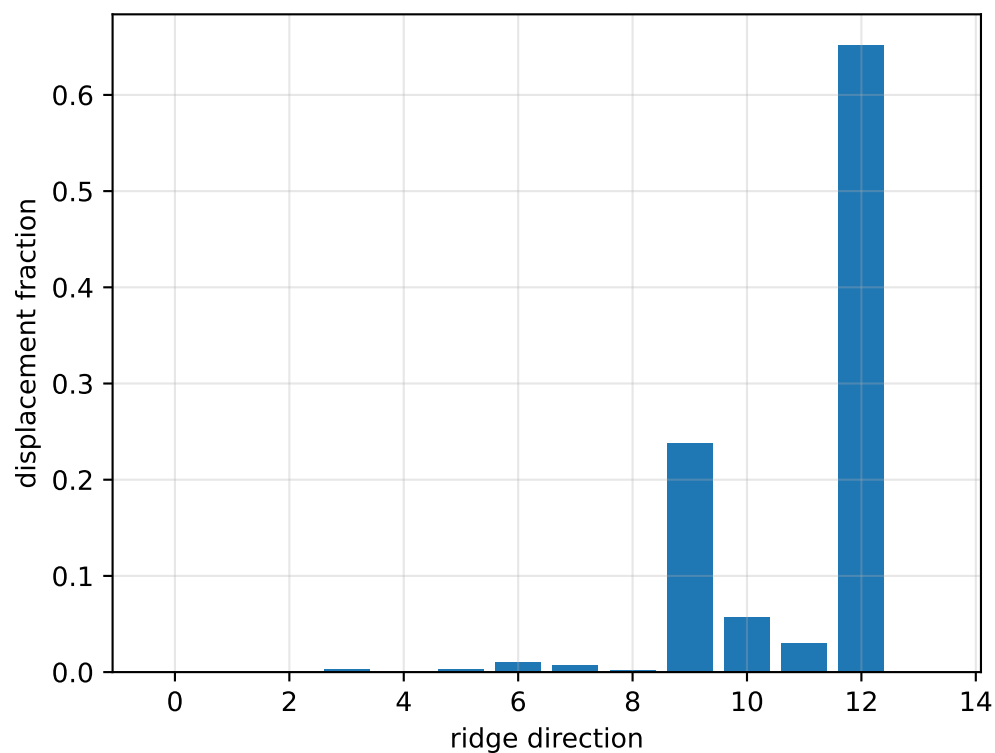
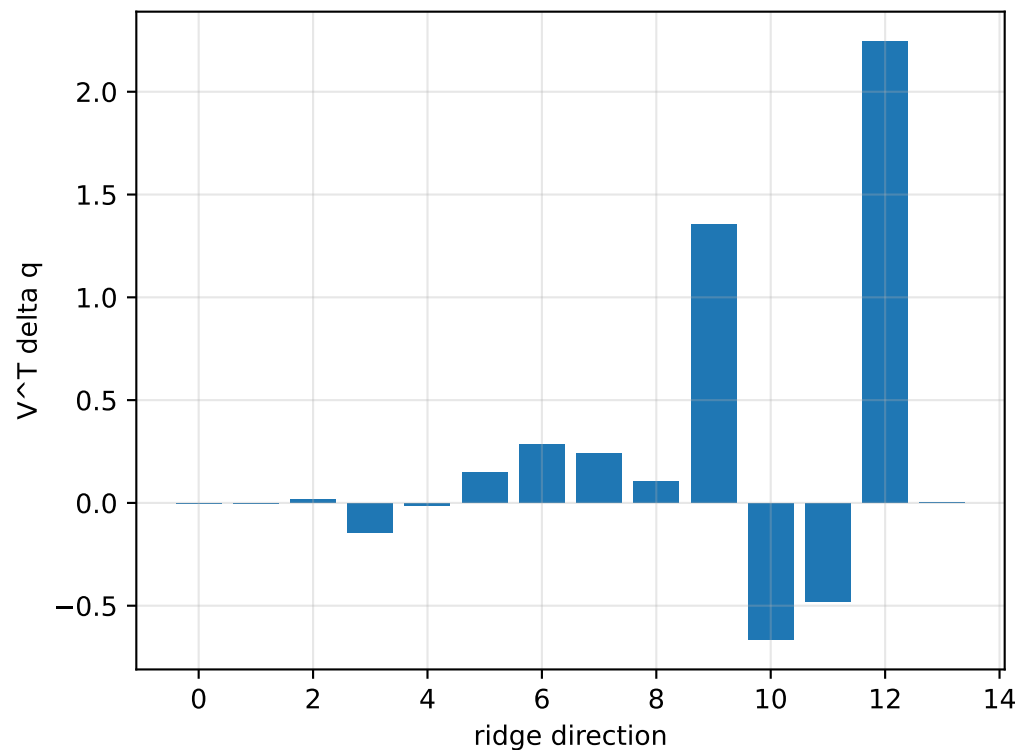
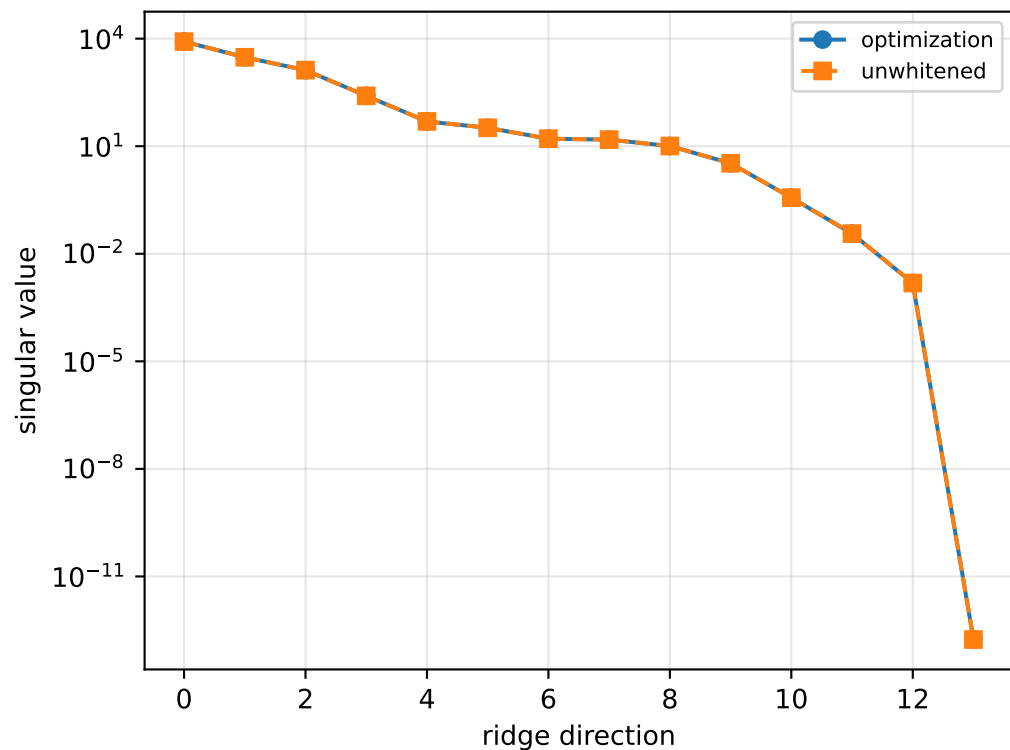
```
[[ 0.07481581  0.00227823 -0.00127729]
 [ 0.00227823  0.02762065 -0.00496393]
 [-0.00127729 -0.00496393  0.10154071]]
```

scale-free f/m: [0.38641993 0.3244967 0.3300864 0.39989412]

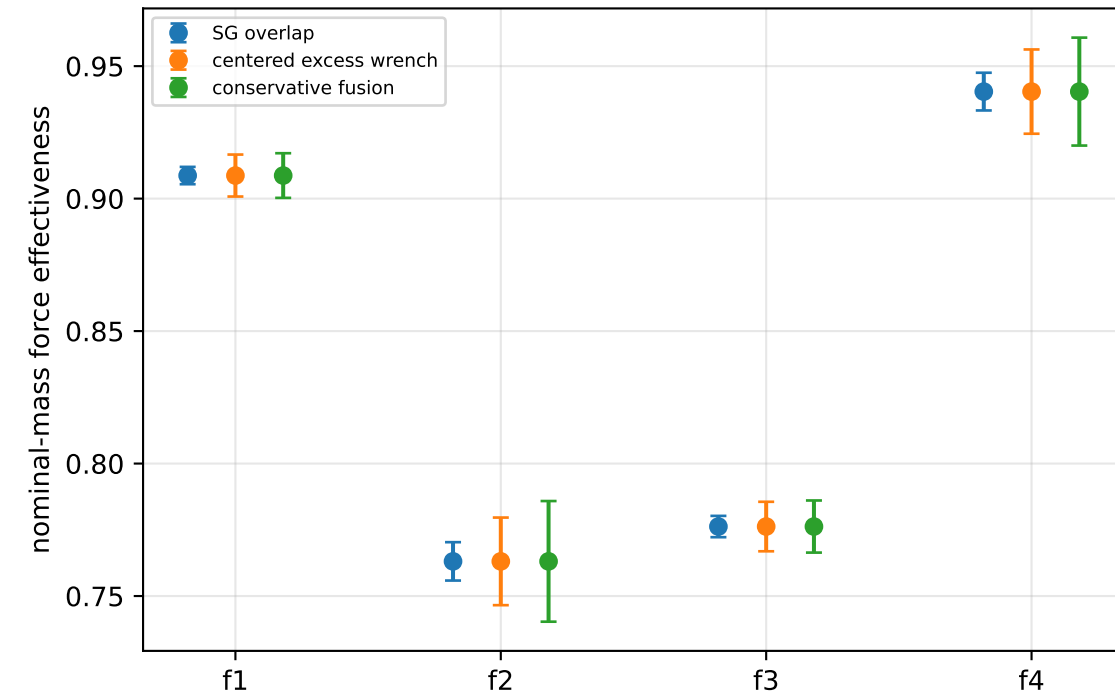
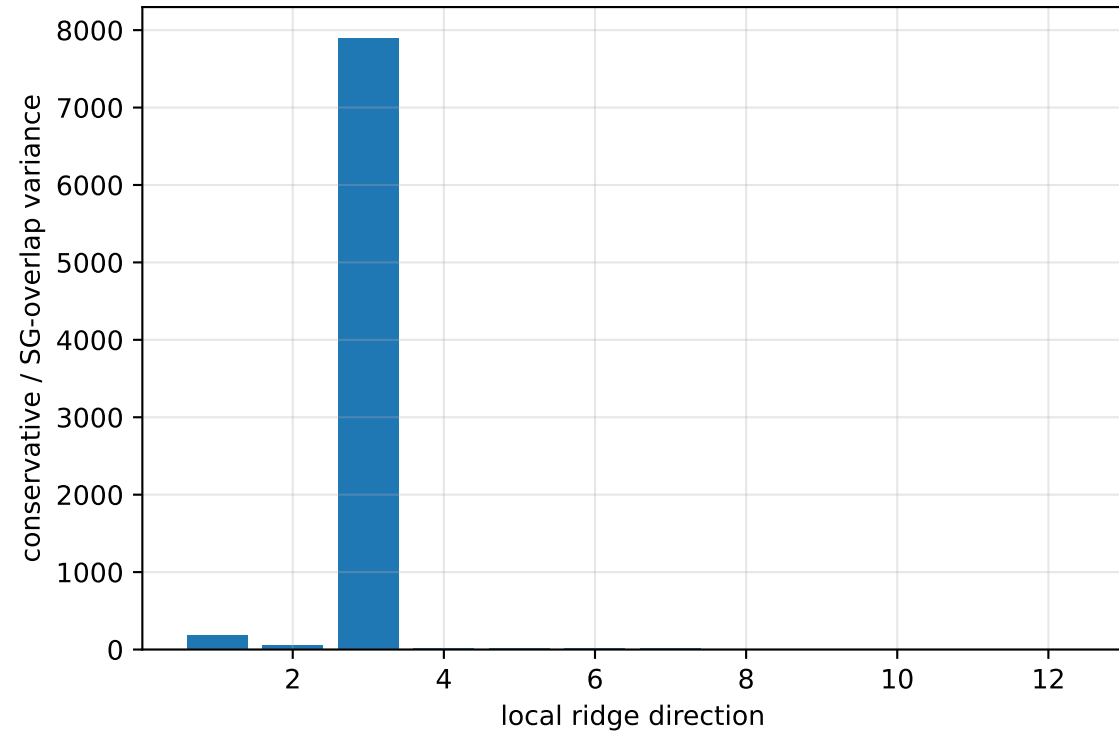
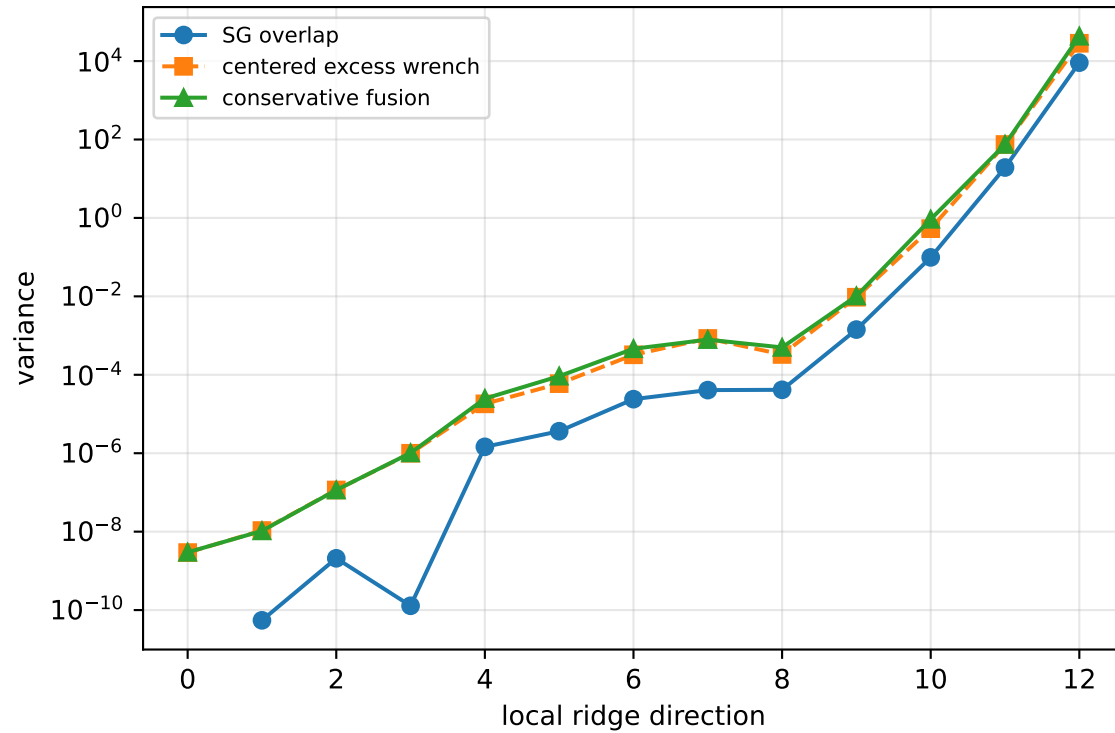
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

inertia_over_mass_yy_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.63e-13)



inertia_over_mass_yy_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 241.3$
wrench-to-acceleration recovery max error = $2.78\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

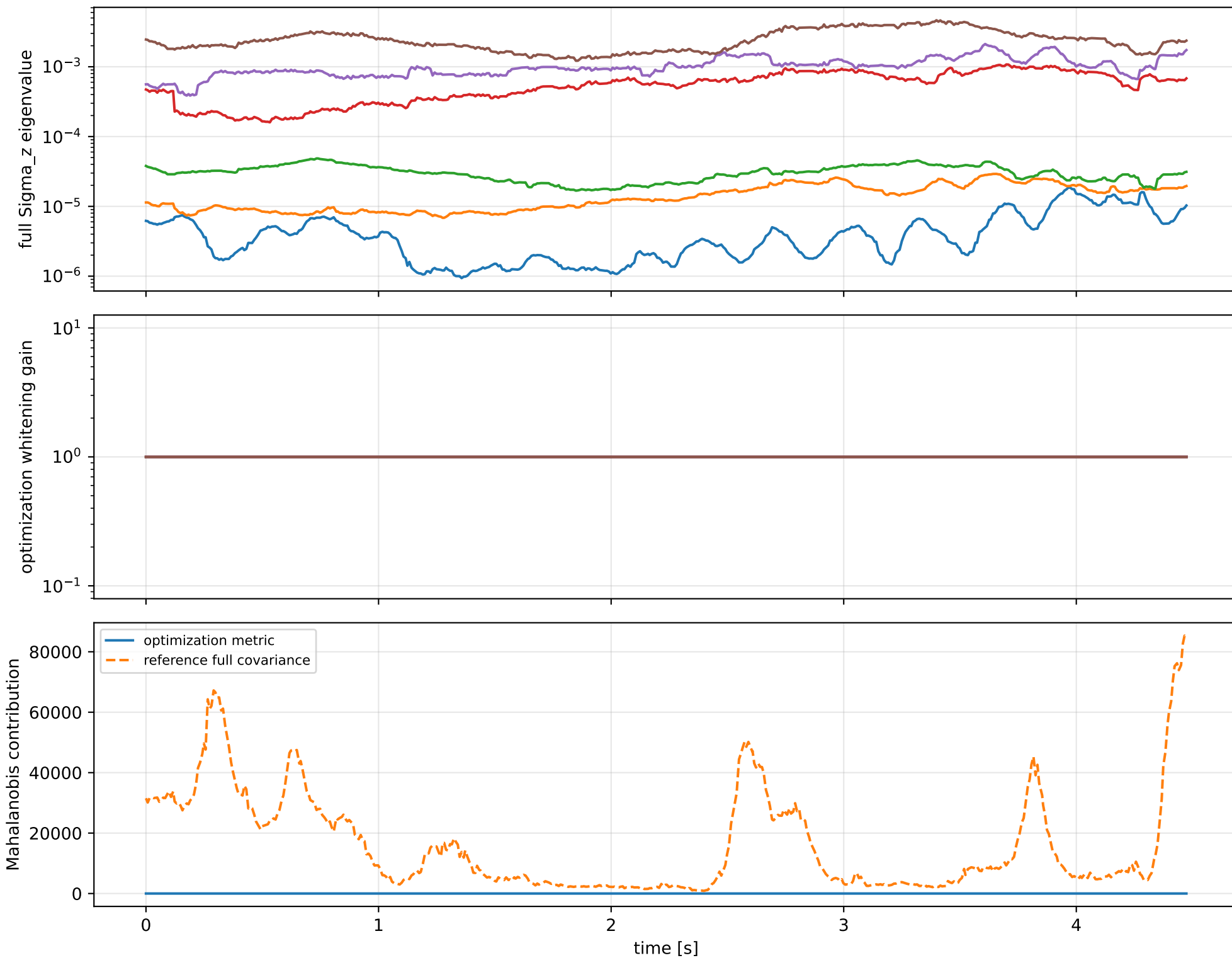
index 12: variance= $4.29\text{e}+04$, ratio= 4.706
 $\log_{\text{mass}}=+0.136$; second_moment_diag_1=-0.0156
second_moment_diag_2=-0.172; second_moment_diag_3=-0.494
second_moment_offdiag_12=-0.306; second_moment_offdiag_13=+0.415
second_moment_offdiag_23=+0.606; cog_x=-3.58e-06
cog_y=-1.52e-06; cog_z=-4.33e-10
log_force_eff_1=+0.136; log_force_eff_2=+0.136
log_force_eff_3=+0.136; log_force_eff_4=+0.136

index 11: variance= 74.9 , ratio= 3.884
 $\log_{\text{mass}}=+0.0153$; second_moment_diag_1=+0.415
second_moment_diag_2=-0.099; second_moment_diag_3=-0.392
second_moment_offdiag_12=+0.389; second_moment_offdiag_13=-0.654
second_moment_offdiag_23=+0.29; cog_x=-8.07e-05
cog_y=+2.14e-05; cog_z=-5.75e-06
log_force_eff_1=+0.0156; log_force_eff_2=+0.0148
log_force_eff_3=+0.0151; log_force_eff_4=+0.0158

index 10: variance= 0.9231 , ratio= 9.365
 $\log_{\text{mass}}=+0.0268$; second_moment_diag_1=-0.0101
second_moment_diag_2=+0.527; second_moment_diag_3=-0.654
second_moment_offdiag_12=+0.169; second_moment_offdiag_13=+0.202
second_moment_offdiag_23=-0.469; cog_x=+0.00159
cog_y=-0.00204; cog_z=+0.000191
log_force_eff_1=+0.0275; log_force_eff_2=+0.0465
log_force_eff_3=+0.0235; log_force_eff_4=+0.0139

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_yy_nominal: optimization vs reference covariance



inertia_over_mass_yy_nominal: wrench / closure (force max $7.55\text{e-}15$, torque max $1.25\text{e-}16$)



inertia_over_mass_yy_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_yy_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 62834c577333da807f8b968827a18fec3976eb136afd16ab23bddf481b6a6c3e

data objective: 115.947259218

prior objective: 6.87522751309e-07

total objective: 115.947259906

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_yy_nominal (inertia_over_mass_m2)

components: yy

target: [0.02762065]

std: [1.e-06]

final: [0.02762065]

error: [1.17262334e-09]

standardized residual: [0.00117262]

factor objective: 6.87522751309e-07